Software Requirements Specification (SRS) for Judiciary Information System

1. Introduction

1.1. Purpose

The primary purpose of this Software Requirements Specification (SRS) is to precisely define the functional and non-functional requirements for the **Judiciary Information System (JIS)**.

The **main objective of the JIS** is to modernize and streamline judicial operations for the Attorney General's office by providing a secure, efficient, and transparent digital platform. Specifically, it aims to:

- Efficiently handle active court cases from initial registration through judgment.
- Make past court cases easily and securely accessible to authorized lawyers and judges for reference and guidance.
- Automate data entry and management processes for court registrars.
- Provide tools for effective court scheduling and case tracking.
- Enable specific querying and reporting functionalities for key stakeholders.
- Implement a fair mechanism for cost recovery related to historical case access for lawyers.

1.2. Scope

The JIS will be a web-based application designed to streamline and manage various judicial processes as requested by the Attorney General's office.

In Scope:

- Detailed capture and management of court case information (defendant details, crime, dates, locations, officers).
- Generation of unique Case Identification Numbers (CINs).
- Dynamic management of hearing dates, including vacant slot display, adjournments, and recording of proceedings summaries.
- Recording of judgment summaries and case closure, with data retention for future reference.
- Management of presiding judge, public prosecutor, starting, and expected completion dates for trials.
- Secure, role-based access for judges, lawyers, and registrars.
- Tracking and charging mechanism for lawyers access to historical case details.
- Specific query functionalities for court registrars regarding pending, resolved, and upcoming cases, as well as individual case status.

Out of Scope:

- Direct integration with external financial transaction systems for automated billing.
- Offline access capabilities.
- Mobile-native applications (initially, it will be web-responsive).

1.3. Definitions, Acronyms, and Abbreviations

- **JIS:** Judiciary Information System
- **SRS:** Software Requirements Specification
- **CIN:** Case Identification Number - A unique identifier generated by the system for each court case.
- **Registrar:** Court Registrar/Administrator - The user role responsible for data entry, scheduling, and user account management.

- **Attorney General's Office:** The requesting authority and primary stakeholder for the JIS development.
- **HTTPS:** Hypertext Transfer Protocol Secure
- **Authentication:** Verifying user identity to grant access to the system.
- **Authorization:** Granting specific permissions and access levels to authenticated users based on their role.
- **Case:** A legal proceeding or dispute managed within the JIS.
- **Hearing:** A scheduled session where a court case is presented or discussed.
- **Adjournment:** The postponement or rescheduling of a court hearing.
- **Judgment:** The final decision or verdict in a court case.

1.4. References

- Storyboards for Judiciary Information System (Previous document/discussions)
- Relevant national/local judicial regulations and guidelines.

2. Overall Description

2.1. Product Perspective

The JIS is a standalone web application that will serve as the primary digital platform for managing judicial processes as specified by the Attorney General's office.

2.1.1. System Interfaces

- **Database Management System (DBMS):** The core system responsible for persistence, storage, and retrieval of all JIS operational and historical data.

2.1.2. User Interfaces

The system will feature a responsive web-based UI accessible via standard web browsers. It will be designed for intuitive navigation and ease of use, adhering to modern UI/UX principles. Key UI elements will include:

- Secure Login/Logout screens with distinct interfaces or dashboards based on user roles.
- Dedicated forms for court case data entry and updates.
- Interactive calendar view for scheduling and managing hearings.
- Search and listing screens for pending, resolved, and upcoming cases.
- Detailed case view screens with tabbed or sectioned navigation for comprehensive information (defendant, crime, proceedings, judgment).
- Document upload, viewing, and versioning interfaces.

2.1.3. Software Interfaces

The system will interact with:

Web Browsers: Chrome, Firefox, Edge, Safari (latest stable versions).

Operating Systems: Windows, macOS, Linux (for client-side access).

Technologies:

- **Frontend:** JSP (Java Server Pages)
- **Backend**: Servlets
- **Database**: JDBC (Java Database Connectivity)
- **Styling:** CSS
- **Deployment/Cloud:** Google Cloud Platform (GCP) **(or)** Vercel

2.1.4. Communications Interfaces

The system will communicate over **HTTPS** for secure data transmission. All network communications between clients and servers, and between different system components, will be encrypted to ensure data confidentiality and integrity.

2.1.5. Operations

The system will operate 24/7, with scheduled maintenance windows communicated in advance. It will require continuous monitoring for performance, security, and availability. Robust backup and recovery procedures will be in place to minimize data loss and ensure rapid system restoration in the event of failure.

2.2. Product Functions

The core functions of the JIS are designed to meet the specific needs outlined by the Attorney General's office:

- **Case Information Capture:** Detailed recording of all relevant case data upon initiation, including defendant details, crime specifics, dates, locations, and arresting officer.
- **Case Identification:** Automated generation and assignment of unique Case Identification Numbers (CINs) for each case.
- **Hearing and Scheduling Management:** Facilitating the assignment of hearing dates, displaying vacant slots, managing adjournments (with reasons), and recording summaries of court proceedings after each hearing.
- **Historical Case Access for Judges:** Providing judges with secure and unrestricted browsing capability for past court cases to aid in judgment guidance.
- **Historical Case Access for Lawyers (Chargeable):** Allowing lawyers to browse old cases via keyword search, with a system to track and charge for each access to case details.
- **Query and Reporting Capabilities:** Providing the registrar with specific functionalities to query and generate reports on:
  - Currently pending court cases (sorted by CIN, with specific details).
  - Cases scheduled for hearing on a particular date.
  - Status of any particular case by CIN.
- **Document Management:** Enabling the upload, storage, categorization, secure retrieval, and versioning of all case-related documents.

2.3. User Characteristics

- **Judges:** Highly privileged users needing efficient access to current and historical case details, orders, and comprehensive scheduling views. They require clear presentation of critical information for decision-making and guidance.
- **Lawyers/Attorneys:** Users focused on filing documents, tracking case progress, and accessing specific current case details. Critically, they require the ability to browse historical cases via keyword search, understanding that access to details will be tracked for billing purposes. They expect easy document submission and search.
- **Court Registrars/Administrators:** Users with high access levels, primarily responsible for accurate data entry for new and ongoing cases, managing hearing schedules, recording proceedings and judgments, and administrative tasks including creating and deleting user accounts. They require robust data entry forms, efficient administrative tools, and the ability to perform detailed queries and generate reports.
- **Police:** Police users are responsible for creating and submitting case reports, updating crime and arrest details, and uploading evidence as the first point of contact in the justice system. They require a simple, fast interface because their work is time-sensitive, often performed in the field, and demands accuracy. Their

access is restricted to data entry and case tracking, without permission to modify court assignments or judge/lawyer decisions.

2.4. Constraints

- **Regulatory & Legal Compliance:** The JIS must rigorously comply with all applicable national and local judicial laws, court rules, data privacy regulations (e.g., GDPR, or equivalent local data protection acts), and confidentiality mandates regarding data access, retention, and disclosure.
- **Security:** Adherence to industry best practices for web application security (e.g., OWASP Top 10) is mandatory. This includes robust authentication, authorization, data encryption (in transit and at rest), and protection against common cyber threats. Special attention must be paid to securing lawyer billing information and preventing unauthorized access to historical case details.
- **Performance:** The system must handle concurrent users efficiently without significant degradation in response time. All pages should load within 3 seconds. Search queries for current and historical cases must return results within 2 seconds for expected data volumes (e.g., up to 100,000 cases). Document uploads (up to 50MB) should complete within 10 seconds under optimal network conditions.
- **Data Integrity:** Ensure the accuracy, consistency, reliability, and validity of all stored data, especially critical case details, hearing dates, and lawyer access logs for billing.
- **Technology Stack:** Preference for established, secure, and maintainable technologies. Consideration for open-source alternatives where feasible to optimize licensing costs without compromising security or performance.
- **Billing Logic:** The system must accurately track and count each instance a lawyer views the detailed content of a historical court case for future billing.

2.5. Assumptions and Dependencies

- **Internet Connectivity:** All users (judges, lawyers, registrars, public users) will have stable and sufficient internet access to operate the web-based system effectively.
- **Hardware Availability:** Necessary server infrastructure (computing, storage, networking) and client-side devices will be provisioned and maintained to support the system's operation.
- **User Training:** Adequate training and support will be provided to all judicial staff (judges, lawyers, registrars) to ensure effective system adoption and utilization.
- **Legal Framework for Billing:** A clear legal and administrative framework for charging lawyers for access to old cases will be established outside the scope of this software's development, providing the basis for the billing tracking functionality.

3. Specific Requirements

3.1. Functional Requirements

- **User Management:**

The system SHALL allow users (Registrars, Judges, Lawyers,Police) to register with unique credentials (username/email and password).

  - The system SHALL authenticate users based on their provided credentials.
  - The system SHALL support role-based access control (Registrar, Judge, Lawyer, Police) to restrict functionality and data view.

Case Information Management:

  - For each new case, the system SHALL allow Police to enter case details and JIS automatically generate and assign a unique Case Identification Number (CIN).

The system SHALL allow the Registrar to create a new court case record.

○ The system SHALL allow the Registrar to enter other mandatory case details: name of the presiding judge, name of the public prosecutor, the starting date of the trial, and the expected completion date of the trial.

○ The system SHALL allow the Registrar to update any details of a currently active court case.
○ Upon completion of a court case, the system SHALL allow the Registrar to record a summary of the judgment and close the case.
○ The system SHALL retain all details of closed cases for future reference and historical access.

- **Hearing & Scheduling Management:**
  - The system SHALL allow the Registrar to assign an initial date of hearing for each new case.
  - When assigning a hearing date, the system SHALL display vacant slots on any working day to the Registrar, considering court availability and judge's schedule.
  - The system SHALL allow the Registrar to enter the reason for adjournment and assign a new hearing date when a case is adjourned.
  - After a hearing takes place for a case, the system SHALL allow the Registrar to enter a summary of the court proceedings.
    After a hearing (whether proceedings occurred or not), the system SHALL allow the Registrar to assign a new hearing date if the case is not closed.
    The system SHALL provide a calendar view that displays all scheduled hearings for a given date, week, or month.
- **Case Querying & Reporting (Registrar):**
  The system SHALL allow the Registrar to query all currently pending court cases.
  - In response to the pending cases query, the system SHALL print out the cases sorted by CIN.
  - The system SHALL allow the Registrar to query cases that have been resolved over any given period (date range).
  - The system SHALL allow the Registrar to query cases that are coming up for hearing on a particular date.
  - The system SHALL allow the Registrar to query the current status of any particular case, identified by its CIN.
- **Historical Case Access & Billing:**
  - The system SHALL allow Judges to browse through the old (closed) cases for guidance on their judgment, with unrestricted access to details.
  - The system SHALL allow Lawyers to browse through old (closed) cases to prepare their lines of arguments.
  - The system SHALL charge Lawyers for each time they view the details of a closed court case.
  - The system SHALL provide the Registrar with a report or interface to view the usage count for each lawyer's historical case access.

3.2. Non-Functional Requirements

- **Performance:**
  - The system SHALL load all application pages within 3 seconds under normal load (up to 100 concurrent users).
  - Search queries (for current and historical cases, including keyword searches) SHALL return relevant results within 2 seconds for a database containing up to 500,000 cases.
    Document uploads (up to 50MB) SHALL complete within 10 seconds under optimal network conditions.
- **Security:**
  - The system SHALL implement robust role-based access control mechanisms to ensure users can only access functionalities and data commensurate with their assigned roles.

  - The system SHALL log all critical security events, including but not limited to: failed login attempts, successful logins, unauthorized access attempts, and data modification events.
  - The tracking data for lawyer's paid views SHALL be securely stored and protected from unauthorized modification.
- **Reliability:**
  - The system SHALL have an overall uptime availability of 99.5% during operational hours, excluding scheduled maintenance windows.
  - The system SHALL handle unexpected errors gracefully, displaying informative and user-friendly error messages without exposing sensitive system details or crashing.
- **Usability:**
  - The system User Interface (UI) SHALL be intuitive, consistent, and easy to navigate for all user roles, minimizing the learning curve.
  - New users (Judges, Lawyers, Registrars,Police) SHALL be able to perform their primary tasks with minimal initial training (e.g., less than 2 hours of formal training).
  - The system SHALL provide clear and immediate feedback for all user actions, including success messages, validation errors, and loading indicators.
  - The system SHALL adhere to a consistent visual design and branding throughout, ensuring a professional and coherent user experience.
  - Forms for data entry shall be designed for efficiency, minimizing redundant input and guiding the user.
- **Portability:**
  - The system SHALL be designed to be deployable on various cloud platforms (e.g., AWS, GCP, Azure) or on-premise infrastructure with minimal configuration changes.
  - The client-side application SHALL be compatible with the latest two stable versions of major web browsers (Chrome, Firefox, Edge, Safari).
- **Accessibility:**
  - The system SHALL comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to ensure it is usable by individuals with disabilities. This includes considerations for keyboard navigation, screen reader compatibility, sufficient color contrast, and proper semantic HTML.